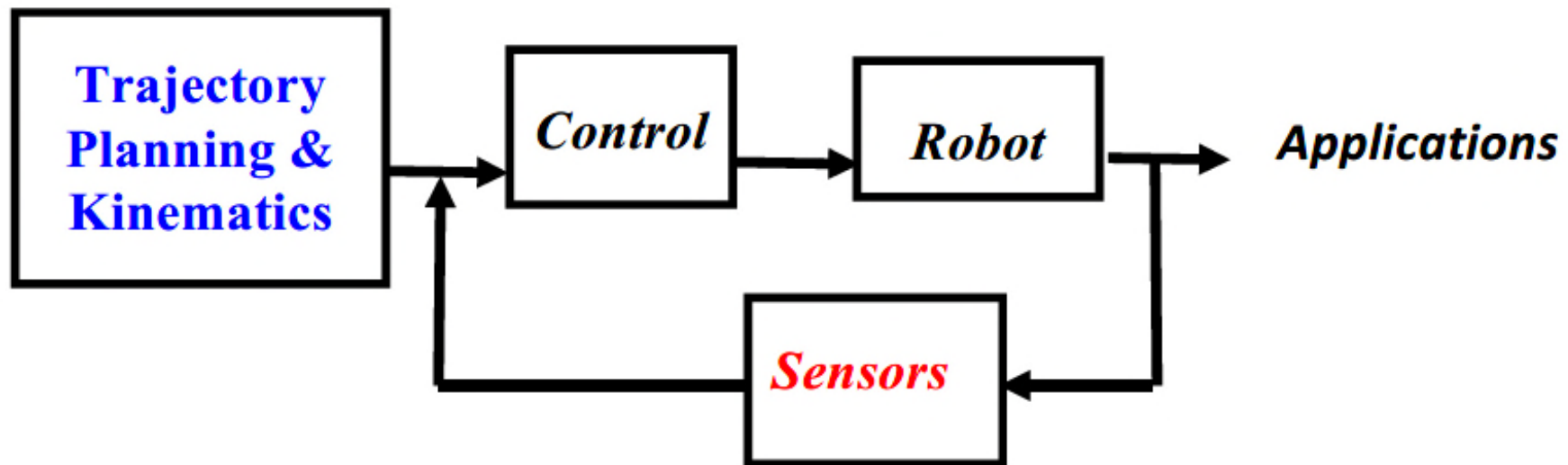


Course Outline

1. **Basics and Applications:** What do we use robot for?
2. **Kinematics (manipulator):** What is the geometric relationship between joints and end effector?
3. **Trajectory Planning:** How to plan a task?
4. **Robot Control:** How to cause the robot to move?
5. **Mobile Robot:** How to model a mobile robot?



REFERENCES

1. Schilling R. J., **Fundamentals of Robotics: Analysis and Control**, Prentice Hall.
2. Craig J. J., **Introduction to Robotics**, Prentice Hall.
3. Siegwart R., Nourbakhsh I.R., and Scaramuzza D., **Introduction to Autonomous Mobile Robots**, MIT press.